

11 (**long**) The owner of a smallholding plans to install a ground-based set of solar panels, and wishes to investigate their sustainability and costs. The owner decides to design and build the frame themselves before having the panels installed professionally.

(a) Considering first the support frame, they assess a range of structural materials without constraining the shape of the members. These will be of fixed length L , with a specified bending stiffness S , to withstand the panel weight and wind loading without excessive deflection. The bending stiffness is given by

$$S = C_1 EI / L^3$$

where C_1 is a constant depending on the support constraints, E is Young's modulus, and I is the second moment of area of the members. For a beam of cross-sectional area A , the shape factor ϕ_e is defined as

$$\phi_e = 12I / A^2.$$

Derive a performance index for the minimum embodied energy H for the frame members, for a specified bending stiffness, when both A and ϕ_e may vary. Rank the materials in Table 1 using this performance index, assuming that each material is available with the maximum shape factor indicated. [8]

(b) The owner contacts the panel installer and learns that their standard mounting system requires the frame to be built using circular sections with a fixed diameter $D = 60$ mm. They recommend using steel tubing of wall thickness $t = 2.5$ mm.

(i) Find the bending stiffness EI of the recommended steel tubing. Subsequently, identify which one of the other three materials can match the steel tubing for stiffness in the same diameter, assuming in the first instance that a solid cross-section will be required. [5]

(ii) Calculate the mass per unit length of the steel tubing, and of the other material options that meet the stiffness requirement in (i), again assuming a solid cross-section is selected. Find the embodied energy per unit length of each viable material. [4]

(iii) All the suitable frame materials are manufactured in the UK, but need to be transported 500 km by truck, for which the transport energy is $1.5 \text{ MJ tonne}^{-1} \text{ km}^{-1}$. For a total length of frame material of 40 m, identify which material from (ii) offers the lowest combination of embodied energy plus transport energy. [3]

(c) The panel array consists of 8 panels, each weighing 18 kg, of which 80% by weight is Si and the rest is equal fractions of Al alloy and glass. The embodied energies of Si and glass are 120 and 39 MJ kg⁻¹, respectively. The panels are shipped 11,500 km from Asia to the UK by sea freight, for which the transport energy is 0.18 MJ tonne⁻¹ km⁻¹, followed by a further distance of 700 km by truck within the UK. Find the total embodied energy and transport energy for the solar panels. [5]

(d) The power output of the panel array averaged over a year is 300 W.

(i) Find the total energy output of the panel array per year, and hence estimate the energy payback period, accounting for the embodied and transport energies of both frame and panels. [3]

(ii) The total installed cost is estimated to be £7000. For an electricity unit cost of £0.40 kWh⁻¹, estimate the cost payback period. [2]

Material	Density ρ (Mg m ⁻³)	Young's Modulus E (GPa)	Embodied energy H_m (MJ kg ⁻¹)	Maximum shape factor ϕ_e
Steel	7.8	210	32	20
Al alloy	2.8	72	190	17
GFRP	1.9	20	120	14
Softwood	0.5	9.3	11	3

Table 1